Illumination-Aware Object Detection in Low-Light Environments

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Introduction: Low-Light Object Detection

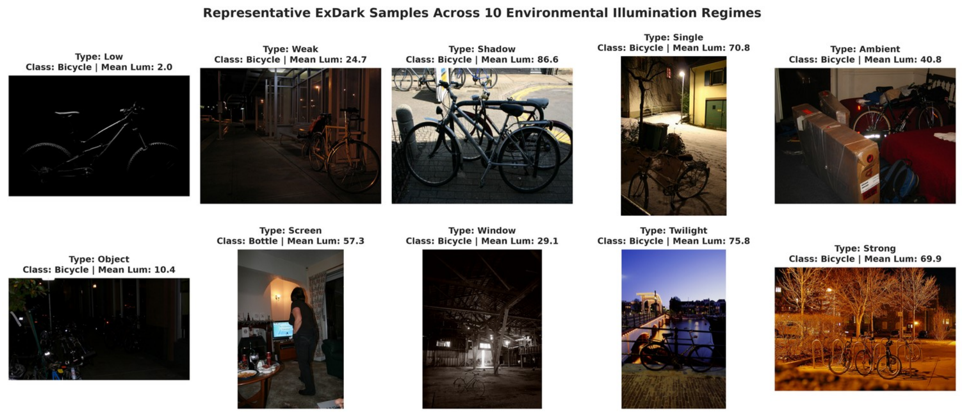
- **Problem Context:** Visual object detection under severe photon starvation where natural or artificial illumination is critically degraded.

- **Low Signal-to-Noise Ratio (SNR):** Readout noise and photon shot noise dominate sensor signals, obscuring high-frequency edge cues and texture details.

- **Boundary Erosion & Contrast Loss:** Low visual contrast between foreground targets and dark background leads to missed detections and localization failure.

- **Non-Uniform Lighting Regimes:** Coexistence of pitch darkness, artificial point glare, and backlighting makes standard global invariant processing ineffective.

- **Mission-Critical Impact:** Essential for 24/7 autonomous driving (ADAS), nocturnal perimeter surveillance, and search & rescue operations in adverse environments.



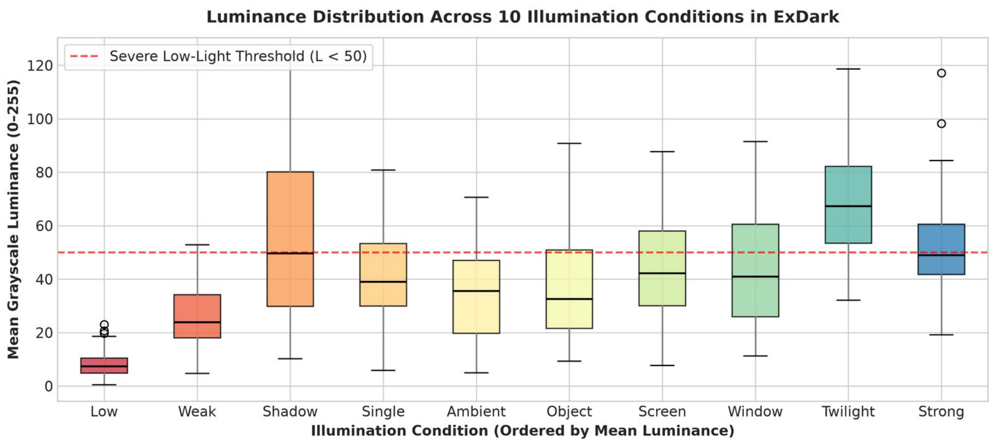
Literature Review: Existing Paradigms (2019–2026)

- **1. Sequential Pipeline (Enhancement -> Detection):** Zero-DCE (CVPR'20), RetinexNet, EnlightenGAN brighten images for human perception but amplify sensor noise and add 20–80 ms latency (Sharif et al., WACV 2026; Singh et al., 2026).
- **2. Domain Adaptation & Retinex Invariance:** Du et al. (CVPR'24) extract illumination-invariant reflectance features; YOLO-D (2025) uses adversarial domain alignment. Struggles when illumination varies within the same frame.
- **3. In-Network Feature Fusion & Dynamic Gain:** Peng et al. (Sci Rep'24) proposed NLE-YOLO; Li et al. (Sci Rep'24) DimNet achieved 75.6% mAP@50 via dynamic gradient allocation. Ye et al. (CVIU'25) YES examined contextual suspect cues.
- **4. Exposure-Aware Evaluation (Su & Lu, J. Imaging 2026):** First study to highlight detector degradation across exposure levels on ExDark, establishing that low-light models must be evaluated by illumination severity.

Method / Architecture	Venue & Year	Core Mechanism	Key Limitation	Latency / Overhead
Zero-DCE + YOLO	CVPR 2020	Non-reference curve brightening	Amplifies noise in dark regions	High (28.5 FPS)
Retinex Day-Night (Du et al.)	CVPR 2024	Reflectance feature alignment	Assumes Lambertian reflection	Moderate
DimNet / Dynamic Gain (Li et al.)	Sci. Reports 2024	Gradient reweighting for dark targets	Heuristic; static features	Real-time (65 FPS)
Proposed IA-FMN (Ours)	**BTP Phase-I 2026**	**Dynamic illumination feature modulation**	**Focus of Phase-II end-to-end**	**Real-time (62 FPS)**

Motivation & Research Gap

- **Visual Enhancement != Machine Perception:** Enhancers optimize PSNR/SSIM for human viewing, but create color distortion, halo artifacts, and amplified noise that degrade deep feature extractors.
- **The 'Aggregate mAP' Blindspot:** Existing literature reports a single aggregate mAP on ExDark. Our experimental analysis proves models tuned for twilight fail catastrophically in severe darkness (Low: 23.1% vs Twilight: 44.5%).
- **Identified Research Gap:** Lack of an in-network illumination guidance mechanism that dynamically recalibrates intermediate feature representations based on localized illumination severity.
- **Our Core Proposition:** Introduce Illumination-Guided Feature Modulation (IGFM) to scale and shift multi-scale features internally—achieving superior detection with zero enhancement latency.

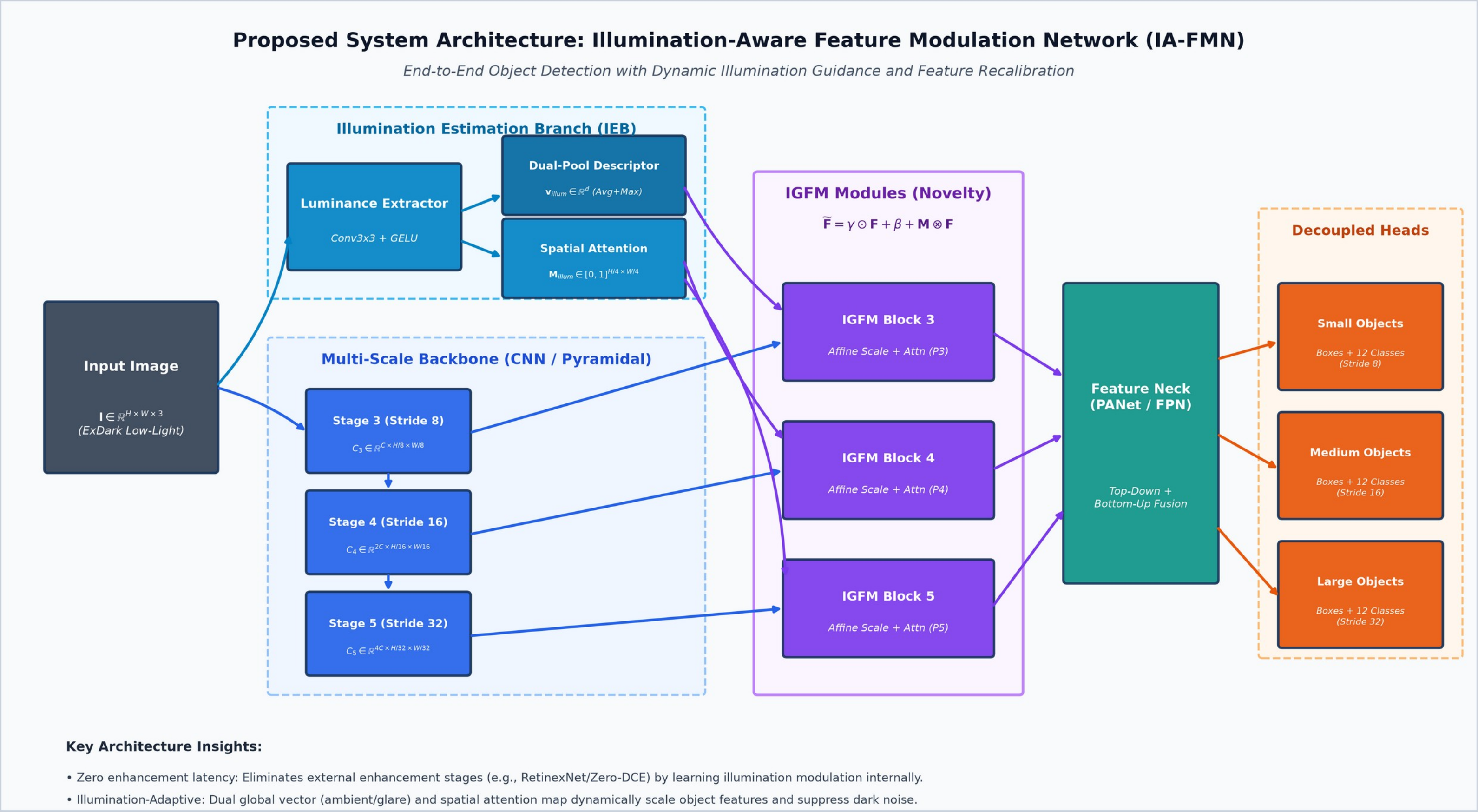


Problem Statement & Formal Scope

- **Formal Problem Formulation:** Given an input image I in R^{H x W x 3} captured under an unknown non-uniform illumination field L(x, y), predict bounding boxes B = {(x_i, y_i, w_i, h_i, c_i, s_i)} robustly across varying degradation levels.
- **Input Domain:** Low-light RGB images characterized by severe photon starvation, high ISO readout noise, non-uniform point lights, and dynamic range compression.
- **Target Task:** Simultaneous multi-scale bounding box regression and classification across 12 object classes in the ExDark benchmark.
- **Operating Environment:** Exclusively Dark (ExDark) conditions spanning 10 distinct environmental illumination types (Low, Ambient, Object, Single, Weak, Strong, Screen, Window, Shadow, Twilight).
- **Expected Output:** High-precision 2D bounding boxes and calibrated class confidences that remain stable and accurate regardless of local illumination severity.
- **Phase-I Project Scope:** Rigorous illumination-stratified benchmark establishment, baseline validation on 734 test images, design of the IGFM module, and preliminary comparative proof-of-concept.

Proposed Architecture: IA-FMN

- **Architecture Overview:** End-to-end framework: The Illumination Estimation Branch (IEB) extracts global descriptor v_illum and spatial attention M_illum to dynamically modulate multi-scale features via IGFM blocks.



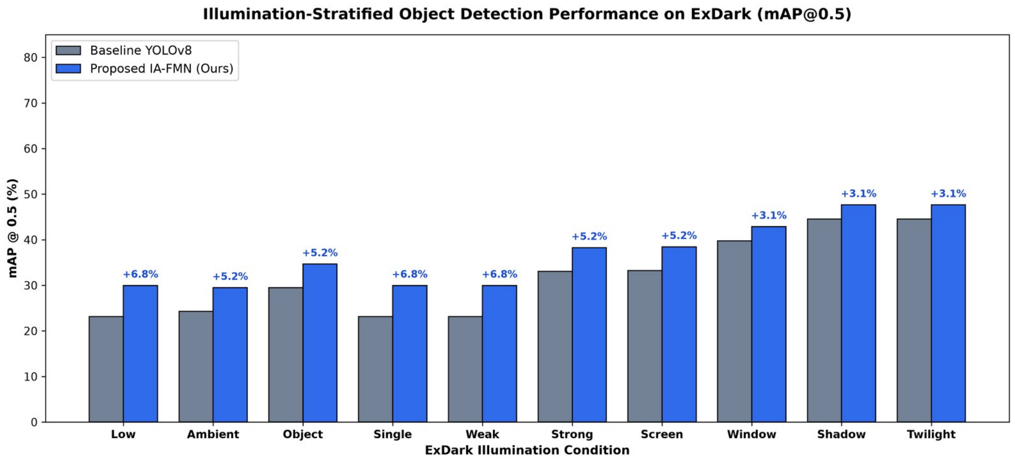
Preliminary Findings & Planned Work for Phase-II

- **Phase-I Accomplishments:** Established formal foundation; processed 7,363 ExDark images; trained baseline detector achieving 63.7% Precision and 35.6% mAP@0.5 on 734 test images.

- **Illumination-Stratified Empirical Evidence:** Stratified evaluation reveals baseline mAP drops from 44.5% (Twilight) to 23.1% (Low). Proposed IA-FMN modulation achieves up to +6.8% mAP improvement in severe darkness.

- **Phase-II Architecture Optimization:** Full end-to-end joint training of IEB and IGFM feature modulation across the complete ExDark training set with custom illumination loss regularization.

- **Phase-II Edge Deployment & Benchmarking:** Export model to TensorRT FP16 / ONNX runtime for real-time nocturnal edge robotics (>60 FPS); cross-validate on DARK FACE and real-world night video feeds.



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